

GNANESWAR REDDY CHAPAALA

Backend API Developer

VA, USA | +1(667)-646-8891 | gnaneswar.chapaala2000@gmail.com | [LinkedIn](#)

SUMMARY

- Played a key role in designing, developing, and maintaining **RESTful APIs** for a large-scale project called **PMJAY 2.0** (which is a **health assurance** program) using **Java** and **Spring Boot**, ensuring high performance, security, and scalability.
- Solely managed the **Health Benefit Packages (HBP)** application, responsible for handling the entire master data of the PMJAY 2.0 project, end-to-end development, robust system architecture, and seamless data management for the **PMJAY 2.0** project.
- Integrated **Kafka** and **Redis** in the HBP application to enhance real-time data processing and **in-memory caching**, optimizing event-driven architecture and ensuring high availability and scalability for critical operations.
- Collaborated closely with **DevOps**, **UI/UX**, and **QA** teams to ensure smooth integration of backend systems with frontend applications and streamline deployment processes through CI/CD pipelines using Jenkins and Docker.
- Led quality assurance efforts by conducting unit testing with **JUnit** and integration testing with Mockito, focusing on early issue identification and maintaining high standards of code quality.
- Actively participated in **Agile Scrum frameworks**, effectively managing tasks and tracking progress using **JIRA** and **MINDCRAFT**, contributing to efficient **sprint cycles** and team responsiveness.
- Worked on a **cancer detection** project using AWS technologies like **S3**, **Glue**, **SageMaker**, and **Athena**, developing data pipelines and machine learning models for accurate diagnosis.
- Handled various image processing tasks, including extracting text from images using OCR techniques and integrating the data into downstream applications.
- Developed and optimized **Android applications** using Java and Kotlin, achieving significant improvements in code efficiency, user engagement, and overall app performance.

TECHNICAL SKILLS

Languages	Java, Python, SQL
Databases Technologies	Relational (MySQL, PostgreSQL), NoSQL (MongoDB, Cassandra)
MS Office Technologies	Microsoft Office (Windows, Word, Excel, Power)
Cloud	Azure Data Studio, Hive, AWS S3, GLUE, Sage Maker, Textract
Networking & Infrastructure	Cisco Networking Basics, Network Troubleshooting
Operating Systems	Windows, Linux, Mac
Methodologies	SDLC, Agile, Waterfall, Scrum
Data Analytics & Visualization	Power BI, Data Mining, Data Visualization, ETL
Machine learning	Scikit-learn, Decision Trees, Random Forest, Predictive Modeling, Regression, Classification, Clustering and Time Series Analysis.
Frameworks	Springboot, Hibernate
Software Technologies	kafka (A distributed event streaming platform or message broker), redis (caching/data storage)

PROFESSIONAL EXPERIENCE

GRADUATE ASSISTANT (GRADER) FOR DISCRETE STRUCTURES | UMBC | USA

Sep2024 - present

- **Grading & Assessment:** Responsible for grading assignments, quizzes, and exams, ensuring fair and accurate evaluation of students' understanding of discrete mathematics. Consistently meet deadlines while maintaining high standards.
- **Student Support:** Provide additional instruction on complex topics like **Boolean algebra, set theory, and combinatorics**. Hold regular office hours and review sessions to enhance students' problem-solving skills and understanding.
- **Course Mastery:** Continuously deepen knowledge of topics like **predicate calculus, number theory, and graph theory** through grading and student interactions, enhancing my ability to communicate complex concepts effectively.
- **Collaborative Environment:** Work closely with the course instructor, aligning grading with course objectives and providing feedback on common student challenges to improve teaching approaches.
- **Educational Tools:** Utilize technology to streamline grading, track progress, and communicate with students, fostering continuous improvement in teaching skills and educational practices.

VITG CORP| PROGRAMMER INTERN| USA

Jun2024 - Sept 2024

- **Healthcare Data Pipeline Development:** Worked on a project focused on cancer detection using **AWS services** like S3, Glue, and SageMaker, building data pipelines to improve healthcare analytics and diagnostic accuracy.
- **Automation & Integration:** Implemented **OCR technology** for medical image processing using AWS Textract, enhancing data extraction accuracy and automating downstream integration tasks by 30%.
- **Cloud Infrastructure & Monitoring:** Designed a **scalable AWS infrastructure** to support secure data transfer for healthcare applications, ensuring compliance with industry standards.
- **Data Transformation:** Utilized **AWS Glue** for ETL processes, streamlining **healthcare data workflows** and improving efficiency in healthcare operations.

TATA CONSULTANCY SERVICES| JAVA DEVELOPER| INDIA

Aug2021 - Aug2023

API Development and Integration:

- **Key Role in PMJAY 2.0:** PMJAY 2.0 (**Pradhan Mantri Jan Arogya Yojana**) under the **National Health Authority (NHA)**. I was deeply involved in the development and maintenance of **RESTful APIs** for the PMJAY 2.0 project, which is a significant initiative under the National Health Authority (NHA). Utilizing Java and **Spring Boot**, I focused on creating APIs that were not only high-performing but also secure and scalable to handle the project's extensive requirements.

Health Benefit Packages (HBP) Application

- **Sole Responsibility:** I was the primary developer responsible for the **Health Benefit Packages (HBP)** application, a core component managing the Master Data of the **Transaction Management System (TMS)** within PMJAY 2.0. My role involved end-to-end development, maintenance, and enhancement of this application. I ensured that the system architecture was robust, and the data management was seamless, acting as the main technical point of contact.
- **Integrated Kafka and Redis in the HBP application** to enhance real-time data processing and in-memory caching, optimizing event-driven architecture and ensuring high availability and scalability for critical operations.

Cross-Functional Collaboration

- **Working with Multiple Teams:** I worked closely with a variety of teams, including **DevOps, UI/UX, and QA**, to ensure smooth integration of all software components. My role involved constant communication and coordination to align backend systems with frontend requirements and deployment processes. Additionally, I collaborated with teams like **HCX, HCP, and BIS**, who relied on my APIs to seamlessly integrate data into their systems. I also played an active role in our CI/CD pipeline, using tools like Jenkins and Docker to automate and streamline deployments, which significantly sped up our release cycles.

Quality Assurance

- **Ensuring Quality:** Testing was a big part of my job. I conducted unit tests using **JUnit** and integration tests with Mockito to make sure the APIs were solid and reliable. I was always focused on catching issues early to maintain high standards in our code.

Agile Methodology

- **Working Agile:** We followed **Agile practices**, specifically within a **Scrum framework**. I was part of **two-week sprints**, which kept our development process flexible and responsive. I managed my tasks through **JIRA** and tracked my progress in the **MINDCRAFT** platform, keeping everything organized and transparent.

- **Developed and maintained Android applications** using **Java** and **Kotlin**, achieving a 20% improvement in code efficiency through optimization and adherence to best practices for Android development. This ensured that the applications were robust, scalable, and performed smoothly across different devices.
- **Created a video browsing application** like YouTube, integrating seamlessly with **YouTube’s server/database**. The app featured a mechanism to detect and enforce patent or copyright restrictions. This feature was 95% accurate in identifying restricted content, preventing unauthorized video playback by displaying a dialog box.
- **Integrated RESTful APIs** into the Android applications, utilizing Retrofit to fetch and display real-time data with a 30% reduction in network latency. This integration enhanced the app’s responsiveness and provided users with up-to-date content efficiently.
- **Implemented local databases using SQLite** to manage and persist application-specific data, improving data retrieval speed by 40% through optimized database schema design and **efficient query execution**. This ensured that the app remained functional and responsive, even with large data sets.
- **Collaborated with design teams** to create engaging, user-friendly interfaces, resulting in a 25% increase in user engagement due to the **seamless** and **intuitive user experience**. I focused on responsive **layouts, smooth animations**, and easy navigation to enhance overall satisfaction.
- **Used Git for version control**, contributing to a collaborative development environment with a 50% reduction in integration issues. By tracking code changes effectively, I ensured smooth collaboration and consistent progress throughout the project lifecycle.
- **Conducted extensive testing and debugging** to ensure that the Android applications performed with 99% stability across various devices and **OS versions**. This rigorous testing process led to a 30% decrease in bug reports and improved overall app reliability.
- **Optimized application performance** by implementing efficient coding practices and memory management techniques, resulting in a 35% reduction in app load times and a 20% decrease in memory usage. This optimization significantly enhanced the user experience by providing a faster and more reliable application.

EDUCATION

Master of Science in Computer Science University of Maryland Baltimore County, USA	Aug2023 - current
Bachelor of Technology in Information Technology GMR Institute of Technology, INDIA	Jun2017 - May2021